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CLEANING APPARATUS AND CLEANING METHOD

Abstract

A cleaning apparatus including a cleaning unit configured to clean a cleaning object includes a supporting portion, which is located on a side of a first surface of the cleaning object and is configured to support the cleaning object; and at least one first supply port, from which a rotational fluid is supplied to the cleaning object supported by the supporting portion. The cleaning apparatus is configured to clean the cleaning object while the cleaning object is rotated by the rotational fluid supplied from the at least one first supply port.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

[0001] This application is based upon and claims the benefit of priority from the prior Japanese Patent Application No. 2024-026233 filed on Feb. 26, 2024; the entire contents of which are incorporated herein by reference.

FIELD

[0002] The present disclosure relates to a cleaning apparatus and a method for cleaning a cleaning object.

BACKGROUND

[0003] A processing apparatus for processing workpieces may clean the workpieces before and after the processing. For example, when manufacturing semiconductor devices, each workpiece such as a wafer may be processed through grinding, polishing, cutting, and laser irradiation, and the processed workpiece may be conveyed to a cleaning apparatus (a cleaning unit provided to the processing apparatus) and cleaned thereat to remove swarf that adhered thereto during the processing. Hereinbelow, a workpiece being an object to be cleaned by the cleaning apparatus may be called a cleaning object.

[0004] In order to efficiently clean the entire cleaning object, for example, Japanese Patent Laid-Open Publication No. 2015-041653 discloses a spinner-styled cleaning apparatus, in which a cleaning object is supported rotatably by a spinner and cleaned while being rotated.

SUMMARY

[0005] The spinner-styled cleaning apparatus may easily clean the entire cleaning object without operating cleaning devices (such as cleaning nozzles, cleaning brushes, etc.) complicatedly. On the other hand, for cleaning the cleaning object while the cleaning object is supported on a supporting surface and being rotated, the cleaning apparatus may require a rotating mechanism, including a motor to rotate the supporting surface and a transmission. As such, downsizing of the cleaning apparatus may be restricted by the rotating mechanism.

[0006] The present disclosure is directed to providing a cleaning apparatus and a cleaning method, which enable a cleaning object to be cleaned efficiently in a downsized structure.

[0007] According to an aspect of the present disclosure, a cleaning apparatus including a cleaning unit configured to clean a cleaning object includes a supporting portion, which is located on a side of a first surface of the cleaning object and is configured to support the cleaning object; and at least one first supply port, from which a rotational fluid is supplied to the cleaning object supported by the supporting portion. The cleaning apparatus is configured to clean the cleaning object while the cleaning object is rotated by the rotational fluid supplied from the at least one first supply port.

[0008] Optionally, the supporting portion may be configured to support an outer edge of the cleaning object and include a side portion configured to cover an exterior of the cleaning object supported by the supporting portion. Moreover, optionally, the at least one first supply port may be formed on an inner circumference of the side portion.

[0009] Optionally, the cleaning apparatus may further include at least one second supply port provided on a supporting surface, which is a surface of the supporting portion configured to support the cleaning object thereon. The cleaning unit may be configured to clean the cleaning object in a state where the cleaning object is elevated from the supporting surface by a floater fluid supplied from the at least one second supply port.

[0010] Optionally, the cleaning unit may be configured to clean at least one of the first surface, a second surface opposite to the first surface, or a side surface, of the cleaning object. Moreover, optionally, the cleaning unit may at least include the at least one first supply port and may be configured to clean the cleaning object with the rotational fluid. Moreover, optionally, the cleaning

unit may include, separately from the at least one first supply port, a cleaning device that may be located at a position to face at least one of the first surface of the cleaning object supported by the supporting portion or a second surface opposite to the first surface of the cleaning object, and the cleaning unit may be configured to clean the cleaning object using the cleaning device.
[0011] Optionally, the cleaning apparatus further includes a lid configured to be arranged to face a second surface opposite to the first surface of the cleaning object. Moreover, optionally, the cleaning apparatus may further include at least one third supply port, through which a fluid is supplied to the cleaning object, on a surface of the lid that faces the cleaning object.
[0012] According to another aspect of the present disclosure, a method for cleaning the cleaning object in the cleaning apparatus includes a supporting step, including supporting the cleaning object with the supporting portion; a rotating step, including supplying the rotational fluid from the at least one first supply port to the cleaning object and causing the cleaning object supported by the supporting portion to rotate; and a cleaning step, including cleaning the cleaning object while the cleaning object rotates.
[0013] According to the above cleaning apparatus and the cleaning method, the cleaning object may be rotated by the rotational fluid supplied from the first supply port and may be cleaned while being rotated, thereby enabling cleaning of the cleaning object efficiently in a downsized structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0014] FIG. **1** is a perspective view of supporting units composing a cleaning apparatus according to a first embodiment.
[0015] FIG. **2** is a perspective view of the supporting units supporting a cleaning object in the cleaning apparatus according to the first embodiment.
[0016] FIG. **3** is a cross-sectional view of the supporting units supporting the cleaning object in the cleaning apparatus according to the first embodiment.
[0017] FIG. **4** is a plan view of the cleaning apparatus according to the first embodiment.
[0018] FIG. **5** is a part of a flowchart to illustrate a flow of processes to be performed in the cleaning apparatus according to the first embodiment.
[0019] FIG. **6** is another part of the flowchart to illustrate the flow of the processes to be performed in the cleaning apparatus according to the first embodiment.
[0020] FIG. **7** is a perspective view of supporting units composing a cleaning apparatus according to a second embodiment.
[0021] FIG. **8** is a cross-sectional view of the supporting units supporting the cleaning object in the cleaning apparatus according to the second embodiment.
[0022] FIG. **9** is a cross-sectional view of supporting units supporting the cleaning object in a cleaning apparatus according to a third embodiment.
[0023] FIG. **10** is a plan view of the cleaning apparatus according to the third embodiment.
[0024] FIG. **11** is a cross-sectional view of supporting units supporting the cleaning object in a cleaning apparatus according to a fourth embodiment.
[0025] FIG. **12** is a plan view of a cleaning apparatus according to a fifth embodiment.
[0026] FIG. **13** is a cross-sectional view sectioned along a line A-A indicated in FIG. **12**.
[0027] FIG. **14** is a cross-sectional view of a cleaning apparatus according to a sixth embodiment.
[0028] FIG. **15** is a plan view of the cleaning apparatus according to the sixth embodiment.
DESCRIPTION OF EMBODIMENTS
[0029] With reference to the accompanying drawings, cleaning apparatuses according to embodiments of the present disclosure and a cleaning method using the cleaning apparatuses according to the embodiments will be described. The cleaning apparatus in each embodiment may

clean a disk-shaped cleaning object S. The cleaning apparatus may either be provided as part of a processing apparatus that may process the cleaning object S or may be an apparatus independent from the processing apparatus. The cleaning apparatus may clean a cleaning object S having been processed by the processing apparatus, a cleaning object S before processing, or a cleaning object S before and/or after another fabrication that does not involve processing (inspection, applying or removal of a tape, image-capturing, etc.). The cleaning object S may be, for example, a semiconductor wafer, a package substrate, or an optical device wafer, but the material or the type of the cleaning object S is not limited.

[0030] A Z-axis direction in each of the drawings illustrating the embodiments is a vertical direction of the cleaning apparatus, where a +Z direction side is upward and a −Z direction side is downward. The Z-axis direction is also a thickness direction of the cleaning object S when the cleaning object S is being cleaned in the cleaning apparatus. The cleaning apparatus in each embodiment may, when cleaning the cleaning object S, supply a rotational fluid to the cleaning object S and cause the cleaning object S to rotate by a force of the rotational fluid. In the cleaning apparatus in each embodiment, a direction in which the cleaning object S rotates will be called a rotating direction. The cleaning object S may rotate along a horizontal plane, which is substantially orthogonal to the Z-axis direction.

[0031] FIGS. **1** through **4** illustrate a cleaning apparatus **10** according to a first embodiment. The cleaning apparatus **10** includes supporting units **11** to support the cleaning object S and a cleaning unit **12** (see FIG. **3**) to clean the cleaning object S supported by the supporting units **11**. The cleaning object S has a first surface Sa, by which the cleaning object S is supported on the supporting units **11**, a second surface Sb on a side opposite to the first surface Sa, and a side surface Sc connecting the first surface Sa and the second surface Sb on an outer circumference. When the cleaning object S is supported on the supporting units **11**, the first surface Sa is on a lower side, and the second surface Sb is on an upper side.

[0032] The supporting units **11** form a ring to support an outer edge of the cleaning object S. The supporting units **11** include an opening portion **19** at a center thereof formed there-through in the vertical direction, and a central portion of the cleaning object S is not directly supported by the supporting units **11**. For example, when the cleaning object S is a semiconductor wafer or an optical device wafer, a central portion of the cleaning object S, on which a device may be formed, is located to coincide with the opening portion **19** in the supporting units **11** so that merely the outer edge of the cleaning object S is supported by the supporting units **11**.

[0033] At positions on the circumference of the supporting units **11**, communicating portions **13** are formed. Two communicating portions **13** are located substantially symmetrically across the center of the ring-shaped supporting units **11**. The communicating portions **13** are each formed as a space to connect the opening portion **19** on an inner side and an outer side of the supporting units **11** in a radial direction. A main function of the communicating portions **13** is to discharge fluid that may flow in the opening portion **19** to the outside of the supporting units **11**; therefore, the configuration of the communicating portions **13** may not necessarily be limited as long as the discharging function is ensured.

[0034] For example, FIGS. **1** and **2** show the supporting units **11**, which are provided in two pieces separated completely by the two communicating portions **13** in the circumferential direction, but optionally, a member in an integrally formed structure, which is continuous in the communicating portions **13** at least partly in the circumferential direction, may compose a supporting unit **11**.

[0035] Each supporting unit **11** includes a supporting portion **14** and a side portion **15**, which are formed integrally. The supporting portions **14** in the supporting units **11** form a ring located on the inner circumferential side of the supporting units **11**, and a thickness thereof in the vertical direction is smaller than the side portions **15**. Each supporting portion **14** has a supporting surface **16**, which is a flat annular surface facing upward. Moreover, on the inner circumferential side of the supporting surface **16**, a discharging portion **17** in a stepped shape is formed to be lower than the supporting surface **16**. The part of the supporting units **11** on the inner circumferential side of the supporting portions **14** forms the opening portion **19**, which is open in the vertical direction. The discharging portions **17** are not required but may be omitted optionally.

[0036] The side portions **15** form a ring located on the outer circumferential side of the supporting units **11**. A thickness thereof of the side portions **15** in the vertical direction is greater than the supporting portions **14** so that the side portions **15** protrude to be higher than the supporting surfaces **16**. The side portions **15** have cylindrically-formed inner circumferential surfaces **18** extending upward from outer edges of the supporting surfaces **16**.

[0037] As shown in FIGS. **2** and **3**, the cleaning object S conveyed to the cleaning apparatus **10** is supported by the supporting units **11**. In particular, the cleaning object S is supported on the annular step formed of the supporting surfaces **16** of the supporting portions **14** and the inner circumferential surfaces **18** of the side portions **15**. The supporting portions **14** are located on a side of the first surface Sa of the cleaning object S, and the annular supporting surfaces **16** may support the outer edge (an area on the first surface Sa closer to the side surface Sc) of the cleaning object S. The side portions **15** are located to cover an exterior of the side surface Sc of the cleaning object S Sc of the cleaning object S, and the inner circumferential surfaces **18** in the cylindrical form, of which inner diameter is greater than a diameter of the cleaning object S, faces the side surface Sc.

[0038] The side portions **15** each have a plurality of first supply ports **20** formed to open on the inner circumferential surfaces **18**. The plurality of first supply ports **20** are located at predetermined intervals along the circumferential direction of the supporting units **11** (in the rotating direction of the cleaning object S). The plurality of first supply ports **20** are located at a position in the vertical direction on the inner circumferential surfaces **18** toward the supporting surfaces **16** closer to a lower end of the inner circumferential surfaces **18**. The plurality of first supply ports **20** arranged in this manner are located to face the side surface Sc of the cleaning object S supported by the supporting units **11**.

[0039] As shown in FIGS. **3** and **4**, inside each supporting unit **11**, a fluid supply path **21** extending in the circumferential direction is provided. The fluid supply path **21** is a flow path located annularly on the outer side of the inner circumferential surface **18** and is formed in a shape of an arc with a curvature radius being greater than a curvature radius of the inner circumferential surface **18**. The plurality of first supply ports **20** are formed as flow paths branching off from the fluid supply path **21** and extending to the inner circumferential surface **18**. As shown in FIG. **4**, each first supply port **20** has a form, in a plan view, that changes circumferential position thereof as it extends away from the fluid supply path **21** and approach the inner circumferential surface **18**. In other words, the first supply ports **20** each have a flow path in a shape including a component of the rotating direction of the cleaning object S supported by the supporting units **11** (which is non-parallel to the radial direction of the cleaning object S).

[0040] The cleaning apparatus **10** is provided with a fluid supply source **22**, which may supply fluid to the fluid supply paths **21** and is connected to the fluid supply paths **21** through an open/close valve. The fluid supply source **22** includes a tank to store the fluid and a pump to pump the fluid from the tank. The fluid to be supplied by the fluid supply source **22** is at least one of a liquid, such as pure water, pure water containing agents such as surfactants, or a mixture of two fluids containing air and water, or a gas such as air. In consideration of ease of availability, operating cost, and environment impact after use, it is preferable that the fluid to be supplied by the fluid supply source **22** may be water or air. In a case where the cleaning object S is made of a material or is of a type that is inappropriate to be exposed to water or air, a fluid other than water or air may supplied from the fluid supply source **22**.

[0041] In a case where the fluid supply source **22** is capable of supplying both a liquid and a gas, the fluid supply source **22** may be configured to have a liquid supply source and a gas supply source, and the type of the fluid to be supplied to the fluid supply paths **21** may be selected by an operation of an open/close valve in a flow path connecting to the liquid supply source and the gas supply source. Optionally, a mixed fluid (two fluids) containing both a liquid and a gas may be supplied from the fluid supply source **22**.

[0042] On one of end faces of each supporting unit **11** in the circumferential direction facing the communicating portion **13**, an inlet **211**, from which the fluid supplied from the fluid supply source **22** may be drawn into the fluid supply path **21**, is formed, and on the other of the end faces of each supporting unit **11** in the circumferential direction facing the communicating portion **13**, an outlet **212**, from which the fluid flowed through the fluid supply path **21** may be discharged, is formed.

[0043] The fluid supplied from the fluid supply source **22** to the inlet **211** of each fluid supply path **21** may travel in a flowing direction Ta (see FIG. **4**) inside the fluid supply path **21** in the arc shape in the plan view. The plurality of first supply ports **20** are formed at points to branch from the fluid supply path **21** in an orientation that forms an acute angle with a tangent line of the flowing direction Ta, so that the fluid flowing through the inside of the fluid supply path **21** may easily enter the individual first supply ports **20**. The fluid entering the first supply ports **20** may be jetted out from outlets of the first supply ports **20** formed on the inner circumferential surface **18**.

[0044] The fluid jetted from the first supply ports **20** will be herein called a rotational fluid Fa. In FIGS. **3** and **4**, the rotational fluid Fa is represented schematically in arrows. The rotational fluid Fa jetted along the shape of the first supply port **20** is oriented in a direction including a component of the rotating direction of the cleaning object S. In particular, the rotating direction of the cleaning object S is substantially the same as the flowing direction Ta of the fluid flowing along the fluid supply path **21**. The orientation of the rotational fluid Fa is a composite of a component along the flowing direction Ta and a radial component of the direction from the fluid supply path **21** toward the center (toward the opening portion **19**) of the supporting units **11**. Accordingly, the rotational fluid Fa flows, in a plan view, in the rotating direction of the cleaning object S while skewing toward the center of the supporting units **11**.

[0045] The rotational fluid Fa jetted from the first supply ports **20** may hit the side surface Sc of the cleaning object S that faces the first supply ports **20** and the outer edges of the first surface Sa and the second surface Sb adjacent to the side surface Sc, thereby applying a force in the rotating direction to the cleaning object S. As a result, the cleaning object S supported by the supporting units **11** may be rotated by the force of the rotational fluid Fa.

[0046] The plurality of first supply ports **20** are provided at the predetermined intervals in the rotating direction (circumferential direction of the supporting units **11**) on the side portions **15** covering the exterior of the cleaning object S supported by the supporting portions **14**. With this configuration, when the rotational fluid Fa is jetted from the plurality of first supply ports **20**, the rotational fluid Fa may apply the force in the rotating direction and the force to push toward the center of the radial direction to the cleaning object S. Accordingly, the cleaning object S may be rotated while the position of the cleaning object S in the horizontal direction is stabilized.

[0047] As such, the rotational fluid Fa jetted from the first supply ports **20** may hit the areas in the vicinity of the outer edge of the cleaning object S and cause the cleaning object S to rotate. A flowing speed and a flowing amount of the rotational fluid Fa may be set according to a pressure and amount of the fluid supplied from the fluid supply source **22**. Therefore, with a controller (not shown) in the cleaning apparatus **10** controlling behaviors of the fluid supply source **22**, a rotational speed of the cleaning object S may be set preferably.

[0048] Meanwhile, behaviors of the cleaning apparatus **10** may, as described further below, differ depending on the type of the fluid supplied from the fluid supply source **22**. For example, in a case where the rotational fluid Fa is also used as cleaning water in a cleaning sequence described below, the fluid supply source **22** may be configured to be capable of supplying the fluid that functions as the cleaning water. For another example, when the cleaning apparatus **10** conducts a drying sequence, which will be described further below, the fluid supply source **22** may be configured to be capable of supplying at least a gas so as not to hinder the cleaning object S from drying.

[0049] Moreover, the rotational fluid Fa, when it is liquid rather than gas, is more viscous, and it is easier to apply the rotational force to the cleaning object S. Therefore, a liquid and a gas may be used occasionally according to the rotational force required to the rotational fluid Fa.

[0050] Note that, in the configuration shown in the drawings, the inlet **211** for introducing the fluid supplied from the fluid supply source **22** into the fluid supply path **21** and the outlet **212** for discharging the fluid from the fluid supply path **21** are formed on one and the other end faces of each supporting unit **11**, respectively, in the circumferential direction facing the communicating portion **13**; however, the configuration of the inlet and the outlet of the fluid supply path **21** is not necessarily limited. For example, the inlet and/or the outlet of the fluid supply path **21** may be provided on an outer circumferential surface, an inner circumferential surface, an upper surface, or a lower surface of the supporting unit **11**. For another example, the fluid supply path **21** may have only the first supply ports **20** as the exit of the fluid from the fluid supply path **21** (where no other outlet than the first supply ports **20** is provided in the fluid supply path **21**) so that all of the fluid that flows through the fluid supply path **21** may be jetted from the openings of the fluid supply path **21**.

[0051] As described above, the cleaning apparatus **10** does not rotate the supporting units **11** but rotates the cleaning object S with the rotational fluid Fa supplied through the supporting units **11**. Therefore, a rotating mechanism to rotate the supporting units **11** is not required, and the cleaning apparatus **10** may be provided in a compact and simple structure.

[0052] The supporting portion **14** in each supporting unit **11** has a plurality of second supply ports **23** that are open on the supporting surface **16**. The plurality of second supply ports **23** are provided at predetermined intervals in the circumferential direction of the supporting units **11** (in the rotating direction of the cleaning object S). The plurality of second supply ports **23** in this arrangement are located to face the first surface Sa in the vicinity of the outer edge of the cleaning object S supported by the supporting units **11**.

[0053] As shown in FIG. **3**, inside each supporting unit **11**, a fluid supply path **24** extending in the circumferential direction is formed. The fluid supply path **24** forms an annular flow path located on the inner side in the circumferential direction with respect to the inner circumferential surface **18** of the side portion **15**, having an arc form, of which curvature radius is smaller than the curvature radius of the fluid supply path **21**. The plurality of second supply ports **23** are formed to branch off from the fluid supply path **24** and extend to the supporting surface **16**. As shown in FIG. **3**, the second supply ports **23** extend substantially vertically.

[0054] The cleaning apparatus **10** is provided with a fluid supply source **25** to supply the fluid to the fluid supply paths **24** and is connected to the fluid supply paths **24** via an open/close valve. The fluid supply source **25** includes a tank to store the fluid and a pump to pump the fluid from the tank. The fluid to be supplied by the fluid supply source **25** is at least one of a liquid or a gas, preferably water, when the fluid is a liquid, or air, when the fluid is a gas. Optionally, depending on the material or the type of the cleaning object S, a fluid other than water or air may supplied from the fluid supply source **25**.

[0055] In a case where the fluid supply source **25** is capable of supplying both a liquid and a gas, the fluid supply source **25** may be configured to have a liquid supply source and a gas supply source, and the type of the fluid to be supplied to the fluid supply paths **24** may be selected by an operation of an open/close valve in a flow path connecting to the liquid supply source and the gas supply source. Optionally, a mixed fluid (two fluids) containing both a liquid and a gas may be supplied from the fluid supply source **25**.

[0056] On one of the end faces of each supporting unit **11** in the circumferential direction facing the communicating portion **13**, an inlet **241**, from which the fluid supplied from the fluid supply source **25** may be drawn into the fluid supply path **24**, is formed, and on the other of the end faces of each supporting unit **11** in the circumferential direction facing the communicating portion **13**, an outlet **242**, from which the fluid flowed through the fluid supply path **24** may be discharged, is formed.

[0057] The fluid supplied from the fluid supply source **25** to the inlet **241** of each fluid supply paths **24** may travel inside the fluid supply path **24** in the arc shape in the plan view. The fluid flowing through the fluid supply paths **24** may enter the individual second supply ports **23** and may be jetted out from outlets of the second supply ports **23** formed on the supporting surface **16**.

[0058] The fluid jetted from the second supply ports **23** will be herein called a floater fluid Fb. In FIG. **3**, the floater fluid Fb is represented schematically in an arrow. The floater fluid Fb jetted from the second supply ports **23**, which extend in the vertical direction, flows upward from the supporting surfaces **16** and hits the first surface Sa of the cleaning object S, applying a force to elevate the cleaning object S. The force of the floater fluid Fb may elevate the cleaning object S from the supporting surfaces **16** and cause the cleaning object S to rotate without contacting the supporting surfaces **16**.

[0059] When the cleaning object S is elevated by the floater fluid Fb and shifts to a state not contacting the supporting surface **16**, a frictional resistance is reduced compared to the case where the first surface Sa of the cleaning object S is in contact with the supporting surfaces **16**, and the cleaning object S may be rotated at a high speed by a low load. Moreover, the higher rotation speed of the cleaning object S during cleaning may improve the cleaning effect. When the frictional resistance against the rotation of the cleaning object S is smaller, the cleaning object S may be rotated efficiently with a relatively small amount of flow of the rotational fluid Fa, and power consumption by the pump of the fluid supply source **22** for supplying the rotational fluid Fa may be reduced.

[0060] Moreover, in the arrangement where the cleaning object S is elevated by the floater fluid Fb and shifted to the state not contacting the supporting surface **16**, the rotating cleaning object S may be prevented from contact with the supporting surfaces **16** by the outer edge thereof and may be prevented from being dirtied or damaged by the supporting surfaces **16**.

[0061] Therefore, by supplying the floater fluid Fb to elevate the cleaning object S, combined effects, such as improving the rotational efficiency of the cleaning object S, improving the cleaning effect by high-speed rotation of the cleaning object S, and preventing stains or damages on the outer edge of the cleaning object S, may be achieved.

[0062] The flowing speed and the flowing rate of the floater fluid Fb are set depending on the pressure or the amount of the fluid to be supplied from the fluid supply source **25**. Therefore, by the controller controlling the cleaning apparatus **10** and the behavior of the fluid supply source **25**, an amount of floatation of the cleaning object S may be varied preferably.

[0063] The behaviors of the cleaning apparatus **10**, as will be described further below, may differ depending on the type of the fluid that may be supplied from the fluid supply source **25**. For example, in a case where the floater fluid Fb is also used as cleaning water in a cleaning sequence described below, the fluid supply source **25** may be configured to be capable of supplying a fluid that functions as the cleaning water. For another example, when the cleaning apparatus **10** conducts a drying sequence, which will be described further below, the fluid supply source **25** may be configured to be capable of supplying at least a gas so as not to hinder the cleaning object S from drying.

[0064] Note that, in the configuration shown in the drawings, the inlet **241** for introducing the fluid supplied from the fluid supply source **25** into the fluid supply path **24** and the outlet **242** for discharging the fluid from the fluid supply path **24** are formed on one and the other end faces of each supporting unit **11**, respectively, in the circumferential direction facing the communicating portion **13**; however, the configuration of the inlet and the outlet of the fluid supply path **24** is not necessarily limited. For example, the inlet and/or the outlet of the fluid supply path **24** may be provided on an outer circumferential surface, an inner circumferential surface, an upper surface, or a lower surface of the supporting unit **11**. For another example, the fluid supply path **24** may have only the second supply ports **23** as the exit of the fluid in the fluid supply path **24** (where no other outlet than the second supply ports **23** is provided in the fluid supply path **24**) so that all of the fluid that flows through the fluid supply path **24** may be jetted from the fluid supply path **24**.

[0065] The supporting units **11** are the paired parts forming a ring divided in half in the circumferential direction at the pair of communicating portions **13**, each having a configuration for supplying the rotational fluid Fa through the first supply ports **20** to the cleaning object S and a configuration for supplying the floater fluid Fb to the cleaning object S through the second supply ports **23**. Although FIG. **4** shows two separate fluid supply sources **22** for ease of illustration, the actual configuration of the apparatus requires at least one fluid supply source **22** and one fluid supply source **25**. Optionally, a single fluid supply source may be configured to double as the fluid supply source **22** and the fluid supply source **25**. In this arrangement, supplying of the fluid from the same fluid supply source to the fluid supply path **21** or the fluid supply path **24** may be managed through, for example, individual open/close valves.

[0066] As shown in FIG. **3**, the rotational fluid Fa and the floater fluid Fb are supplied to the areas in the vicinity of the outer edge of the cleaning object S covered by the side portions **15**. In particular, the rotational fluid Fa is supplied to the area in the vicinity of the side surface Sc of the cleaning object S, and the floater fluid Fb is supplied to the area in the vicinity of the first surface Sa, closer to the side surface Sc of the cleaning object S. If, for example, the rotational fluid Fa or the floater fluid Fb stays accumulatively in the area near the outer edge of the cleaning object S, the posture and the rotation of the cleaning object S tend to become unstable due to the accumulation of the fluid. This tendency may be particularly strong in a case where the rotational fluid Fa or the floater fluid Fb is liquid, of which viscosity is higher than a gas.

[0067] Each supporting unit **11** has a discharging portion **17**, which forms a larger gap with the cleaning object S than with the supporting surface **16**, at an inward and adjacent position with respect to the supporting surface **16**. With this arrangement, the rotational fluid Fa and the floater fluid Fb supplied to the areas in the vicinity of the outer edge of the cleaning object S may be easily discharged toward the opening portion **19** on the inner circumferential side of the supporting units **11** through the discharging portions **17**. Moreover, the rotational fluid Fa and the floater fluid Fb entering the opening portion **19** may be discharged outside the supporting units **11** through the communicating portions **13**, at which the opening portion **19** and the outside of the supporting units **11** communicate. As such, the rotational fluid Fa and the floater fluid Fb may be prevented from remaining in the areas in the vicinity of the outer edge of the cleaning object S, and the cleaning object S may be stabilized in the posture and rotated accurately.

[0068] As shown in FIG. **3**, the cleaning unit **12** includes a cleaning nozzle **26** located above the supporting units **11**. A cleaning-water supply source **27** and an air supply source **28** are connected to the cleaning nozzle **26** via respective open/close valves. The cleaning nozzle **26** may jet cleaning water supplied from the cleaning-water supply source **27**, air supplied from the air supply source **28**, or a mixed fluid containing cleaning water and air (two fluids) downward. The cleaning nozzle **26** is a cleaning device that may be located at a position to face the second surface Sb of the cleaning object S which is supported by the supporting units **11**, and when the cleaning object S is supported by the supporting units **11**, cleaning water, air, or mixed fluid (two fluids) may be jetted from the cleaning nozzle **26** at the second surface Sb of the cleaning object S, and the cleaning object S may be cleaned with the fluid jetted from the cleaning nozzle **26**.

[0069] The cleaning nozzle **26** is movable in the radial direction of the cleaning object S with a nozzle-driving assembly, which is not shown. By moving the cleaning nozzle **26** in the radial direction of the cleaning object S while rotating the cleaning object S by the effect of the rotational fluid Fa, the fluid, such as cleaning water, air, or mixed fluid (two fluids), jetted from the cleaning nozzle **26** may be delivered to the entire second surface Sb of the cleaning object S.

[0070] Cleaning operations and a method performed with the cleaning apparatus **10** configured as above will be described below. The cleaning operations with the cleaning apparatus **10** to the cleaning object S includes the cleaning sequence shown in the flowchart in FIG. **5** and the drying sequence shown in the flowchart in FIG. **6**, where the cleaning sequence and drying sequence are performed continuously. The cleaning apparatus **10** includes the controller, which may control operations and processes in the cleaning sequence and the drying sequence. In the present embodiment, the controller automatically determines a condition and selects a step to proceed at each point where the flow branches in the cleaning sequence and the drying sequence; however, the selection may be made manually by an operator of the cleaning apparatus **10** who determines the condition.

Cleaning Sequence

[0071] In the cleaning sequence, the cleaning apparatus **10** may select an option for cleaning between cleaning the cleaning object S in a state elevated by the floater fluid Fb and cleaning the cleaning object S in a state not being elevated by the floater fluid Fb. Cleaning the cleaning object S in the state not being elevated by the floater fluid Fb is performed by rotating the cleaning object S in a state where the outer edge of the first surface Sa is in contact with the supporting surfaces **16** of the supporting units **11**. As described above, it is preferable that the floater fluid Fb is supplied to the cleaning object S when the cleaning object S is being rotated; however, an option not to supply the floater fluid Fb to the cleaning object S may be available when, for example, frictional resistance between the cleaning object S and the supporting units **11** is small even without the floater fluid Fb.

[0072] In Step **100**, the controller determines an operating condition of the cleaning apparatus **10** in the cleaning sequence. If the operating condition meets the condition for cleaning without using the floater fluid Fb (NO in Step **100**), the control proceeds to Step **101**. In Step **101**, the cleaning object S is conveyed to the supporting units **11** and placed on the supporting surfaces **16** by the conveyer assembly (not shown) without having the floater fluid Fb supplied from the second supply ports **23**. The conveyed cleaning object S is supported on the supporting surfaces **16** by the outer edge of the first surface Sa.

[0073] Optionally, unlike the cleaning apparatus **10** in the present embodiment, a cleaning apparatus not equipped with the function to supply the floater fluid Fb to the cleaning object S may be provided. In this configuration, the control always makes a negative determination NO in Step **100**.

[0074] If the operating condition meets the condition for cleaning with use of the floater fluid Fb (YES in Step **100**), the control proceeds to Step **102**, where the controller determines a setting of timing when the cleaning object S is to be supported by the supporting units **11**.

[0075] In a case where the cleaning object S is to be supported by the supporting units **11** after supplying the floater fluid Fb (NO in Step **102**), the control proceeds to Step **103**, where the floater fluid Fb is supplied from the second supply ports **23**, and thereafter to Step **104**, where the cleaning object S is conveyed by the conveyer assembly to the supporting units **11** and placed on the supporting surfaces **16**.

[0076] In a case where the cleaning object S is to be supported by the supporting units **11** before supplying the floater fluid Fb to the cleaning object S (YES in Step **102**), the control proceeds to Step **105**, where the cleaning object S is conveyed by the conveyer assembly to the supporting units **11** and placed on the supporting surfaces **16**, and thereafter to Step **106**, where the floater fluid Fb is supplied from the second supply ports **23** to the cleaning object S.

[0077] It is preferable that a liquid is the floater fluid Fb to be supplied in Step **103** and Step **106**. If a gas as the floater fluid Fb is supplied for cleaning, the cleaning object S may dry in the vicinity of the area where the cleaning object S is exposed to the jetted floater fluid Fb, causing foreign matter to adhere to the cleaning object S. By using a liquid as the floater fluid Fb, the cleaning object S is wetted, preventing foreign matter from adhering thereto, and the cleaning effect may be improved. Moreover, in the case where the floater fluid Fb is a liquid, the floater fluid Fb itself may clean the cleaning object S. In particular, while the cleaning apparatus **10** has the cleaning unit **12** configured to mainly clean the second surface Sb of the cleaning object S, by supplying the floater fluid Fb being a liquid to the first surface Sa of the cleaning object S, the first surface Sa may be benefited

to be also cleaned to a certain extent.
[0078] The floater fluid Fb supplied in Step **103** or Step **106** may be a mixture (two fluids) of a liquid and a gas. In this case, the same effect may be achieved as the floater fluid Fb being a liquid.
[0079] As the control proceeds to Step **101**, **104**, or **106**, to which the control proceeds through branches in Step **100** or Step **102** according to the respective conditions, the cleaning object S is supported by the supporting units **11** at the outer edge portion thereof as shown in FIGS. **2** and **3**. In other words, Steps **101**, **104**, and **106** are the supporting step for supporting the cleaning object S with the supporting portions **14**. In Step **101**, the cleaning object S is in a state where the first surface Sa is in direct contact with the supporting surfaces **16** of the supporting units **11**. In Step **104** or **106**, the cleaning object S is in a state where the outer edge of the cleaning object S is elevated by the floater fluid Fb to be separated from the supporting surfaces **16**.
[0080] Next, the control proceeds to Step **107**. In Step **107**, the rotational fluid Fa is supplied to the cleaning object S from the first supply ports **20**, causing the cleaning object S to rotate on the supporting units **11** by the effect of the rotational fluid Fa. In other words, Step **107** is the rotating step for supplying the rotational fluid Fa from the first supply ports **20** to the cleaning object S and causing the cleaning object S supported by the supporting portions **14** to rotate. In the cleaning sequence, the cleaning object S is rotated by the rotational fluid Fa supplied from the supporting units **11**. As such, a number of mechanically operable parts in the supporting units **11** is small, and the cleaning apparatus **10** may be provided in a compact and simple structure.
[0081] It is preferable that a liquid is the rotational fluid Fa to be supplied in Step **107**. As in the case of the floater fluid Fb described above, by using a liquid as the rotational fluid Fa, the cleaning object S may be prevented from drying so that foreign material may be prevented from adhering to the cleaning object S, and the cleaning effect may be improved. Moreover, in the case where the rotational fluid Fa is a liquid, the rotational fluid Fa itself may clean the cleaning object S. In particular, while the cleaning apparatus **10** has the cleaning unit **12** located at the position for mainly cleaning the second surface Sb of the cleaning object S, a part of the rotational fluid Fa being a liquid may flow around the side surface Sc to the first surface Sa, and side surface Sc and the first surface Sa may be benefited to be also cleaned to a certain extent.
[0082] Optionally, the rotational fluid Fa to be supplied in Step **107** may be a mixture fluid (two fluids) containing a liquid and a gas. In this case, an effect substantially the same as the effect achievable with the rotational fluid Fa being a liquid may be achieved.
[0083] Once rotation of the cleaning object S is stabilized, the control proceeds to Step **108** to clean the cleaning object S. In other words, Step **108** is the cleaning step for cleaning the cleaning object S while the cleaning object S rotates. Cleaning of the cleaning object S in Step **108** may be performed using the cleaning unit **12** such that, as shown in FIG. **3**, the cleaning unit **12** jets the cleaning water supplied from the cleaning-water supply source **27** downward from the cleaning nozzle **26** and cleans the cleaning object S with the cleaning water jetted from the cleaning nozzle **26**. In the example shown in FIG. **3**, the second surface Sb of the cleaning object S is the main target to be cleaned by the cleaning unit **12**.
[0084] By the cleaning nozzle **26** being movable to the position above the outer edge of the cleaning object S, the cleaning water may be delivered also to the side surface Sc of the cleaning object S and a part of the first surface Sa continuous from the side surface Sc to clean.
[0085] The cleaning unit **12** jetting the cleaning water downward from the cleaning nozzle **26** located above the cleaning object S may apply moderate pressure, which may otherwise cause the cleaning object S to deviate upward from the supporting units **11**, from above to the cleaning object S, thereby preventing the cleaning object S from being elevated excessively. Moreover, it is the cleaning water from the cleaning nozzle **26** that is supplied by the cleaning unit **12**; therefore, the cleaning object S may not be pressed excessively intensely against the supporting surfaces **16** from above.
[0086] For example, unlike the cleaning unit **12** in the present embodiment, if the second surface

Sb of the cleaning object S is cleaned while a tool such as a cleaning brush located above the cleaning object S is urged downward against the second surface Sb of the cleaning object S, resistance against the rotation of the cleaning object S caused by the rotational fluid Fa may increase, and a load to rotate the cleaning object S may increase too large so that it may not be negligible. Moreover, the elevating effect by the floater fluid Fb may be reduced, causing the cleaning object S to slidably rotate on the supporting surfaces **16**, and the outer edge of the cleaning object S may be dirtied or damaged. In contrast, by cleaning the cleaning object S with the cleaning water supplied from the cleaning nozzle **26**, such problems are unlikely to occur.

[0087] As described above, the cleaning object S may be cleaned using the rotational fluid Fa as the cleaning water supplied thereto in Step **107**. In this case, as a first selection in the cleaning process in Step **108**, the cleaning water jetted from the cleaning nozzle **26** and the rotational fluid Fa jetted from the first supply ports **20** may be used in combination for cleaning. As a second selection, without supplying cleaning water from the cleaning nozzle **26**, the rotational fluid Fa supplied from the first supply ports **20** may be jetted as the cleaning water to clean the cleaning object S. In the case of the second selection, the cleaning unit **12** (the cleaning nozzle **26**, the cleaning-water supply source **27**, and the air supply source **28**) may be omitted, and the supporting units **11** may serve to provide the cleaning function of the cleaning unit (the configuration to jet the rotational fluid Fa may serve as the cleaning unit).

[0088] In either the first selection or the second selection, the floater fluid Fb in addition to the rotational fluid Fa may be used as the cleaning water, optionally. By using both the rotational fluid Fa and the floater fluid Fb as the cleaning water, the cleaning effect may be improved even more.

[0089] As such, how the cleaning water may be supplied in the cleaning process in Step **108** may be selected from multiple options. The rotational fluid Fa has been supplied from the first supply ports **20** continuously from Step **107**; therefore, in Step **108**, the controller may merely need to select whether the cleaning water should be supplied from the cleaning nozzle **26** or not, in the configuration where the cleaning apparatus **10** is equipped with the cleaning nozzle **26**. Therefore, complicated control is not required.

[0090] The cleaning process in Step **108** may be terminated on a predetermined time basis or may be ended based on a predetermined condition other than time. The condition other than time may include, for example, a degree of dirtiness removed from the cleaning object S, which may be determined by an amount of dirt contained in the cleaning water being collected or based on information indicating dirtiness of the cleaning object S estimated from, for example, a reflection rate in a captured image of the cleaning object S, so that the controller may continue the cleaning process until the controller determines that the dirtiness is removed from the cleaning object S.

[0091] After the cleaning process in Step **108** is completed, the control proceeds to Step **109**. In Step **109**, whether the rotational fluid Fa is to be supplied continuously or not is selected. For example, in the case where the rotational fluid Fa supplied in Step **107** through Step **108** is a liquid, if the rotational fluid Fa is continuously supplied, the liquid may hinder the cleaning object S from drying in the next drying sequence. Therefore, supply of the rotational fluid Fa may be discontinued. For another example, supply of the rotational fluid Fa may be discontinued uniformly regardless of the type of the rotational fluid Fa once the cleaning process is completed. In these cases, a negative determination NO is made in Step **109**, and the control may proceed to S**110**, where supply of the rotational fluid Fa from the first supply ports **20** is discontinued, and thereafter proceed to Step **111**.

[0092] As described above, it is preferable that the rotational fluid Fa to be supplied in Step **107** through Step **108** is a liquid. However, in a case where the rotational fluid Fa supplied in Step **107** through Step **108** is a gas, it may be determined that the gas does not hinder the cleaning object S from drying in the next drying sequence, and supply of the rotational fluid Fa may be continued during transition from the cleaning sequence to the drying sequence. In this case, a positive determination YES is made in Step **109**, and the control may proceed directly to Step **111** without detouring to Step **110**.

[0093] In Step **111**, whether the floater fluid Fb is to be supplied continuously or not is selected. For example, in the case where the floater fluid Fb supplied in Step **103** or Step **106** is a liquid, if the floater fluid Fb is continuously supplied, the liquid may hinder the cleaning object S from drying in the next drying sequence. Therefore, supply of the floater fluid Fb may be discontinued. For another example, supply of the floater fluid Fb may be discontinued uniformly regardless of the type of the floater fluid Fb once the cleaning process is completed. In these cases, the controller may determine negatively NO in Step **111**, and the control may proceed to S**112**, where supply of the floater fluid Fb from the second supply ports **23** is discontinued, and thereafter proceed to Step **113** (drying sequence) shown in FIG. **6**.

[0094] As described above, it is preferable that the floater fluid Fb to be supplied in Step **103** or Step **106** is a liquid. However, in a case where the floater fluid Fb supplied in Step **103** or Step **106** is a gas, it may be determined that the gas does not hinder the cleaning object S from drying in the next drying sequence, and supply of the floater fluid Fb may be continued during transition from the cleaning sequence to the drying sequence. In this case, a positive determination YES is made in Step **111**, and the control may proceed directly to Step **113** shown in FIG. **6** without detouring to Step **112**.

[0095] In a case where the floater fluid Fb was not selected to be used in Step **100**, the cleaning sequence has been executed without supplying the floater fluid Fb. Therefore, the control may skip Sep **111** and proceed to Step **113** (drying sequence) shown in FIG. **6**.

Drying Sequence

[0096] Next to the cleaning sequence, the control proceeds to the drying sequence shown in FIG. **6**. Similarly to the cleaning sequence as described above, in the drying sequence, the cleaning apparatus **10** may select an option for drying between drying the cleaning object S in a state elevated by the floater fluid Fb and drying the cleaning object S in a state not being elevated by the floater fluid Fb. Drying the cleaning object S in the state not being elevated by the floater fluid Fb is performed by rotating the cleaning object S in a state where the outer edge of the first surface Sa is in contact with the supporting surfaces **16** of the supporting units **11**.

[0097] In Step **113**, the controller determines an operating condition of the cleaning apparatus **10** in the drying sequence. If the operating condition meets the condition for drying using the floater fluid Fb (YES in Step **113**), the control proceeds to Step **114**, where the controller starts supplying the floater fluid Fb from the second supply ports **23** to the cleaning object S. For the floater fluid Fb to be supplied in Step **114**, a gas, rather than a liquid that may wet the cleaning object S, is selected. The control proceeds to Step **115** with supplying the floater fluid Fb to the cleaning object S from the second supply ports **23**. With the floater fluid Fb being supplied, the cleaning object S may float from the supporting surfaces **16**.

[0098] If the operating condition meets the condition for drying without using the floater fluid Fb (NO in Step **113**), the control proceeds to Step **115** without supplying the floater fluid Fb to the cleaning object S from the second supply ports **23**.

[0099] Optionally, in a case where the cleaning apparatus **10** is configured to supply solely a liquid of floater fluid Fb, a negative selection NO may be made automatically in Step **113**.

[0100] In the case where the selection for supplying the floater fluid Fb continuously was made earlier in Step **111**, the floater fluid Fb is being supplied currently (the cleaning object S is floating from the supporting surfaces **16** by the floater fluid Fb having been supplied). Therefore, the control may skip Sep **113** and proceed to Step **115**.

[0101] In Step **115**, the controller determines a supplying status (whether the rotational fluid Fa is being supplied or not being supplied) of the rotational fluid Fa. In a case where supply of the rotational fluid Fa was discontinued in Step **110**, the rotational fluid Fa is not being supplied, and, a positive determination YES is made, in Step **115**, and the control proceeds to Step **116**, where the controller starts supplying the rotational fluid Fa to the cleaning object S from the first supply ports

20. For the rotational fluid Fa to be supplied in Step **116**, a gas, rather than a liquid that may wet the cleaning object S, is selected. With the rotational fluid Fa being supplied, the cleaning object S may be rotated by the effect of the rotational fluid Fa. When the rotation of the cleaning object S is stabilized, the control proceeds to Step **117**.

[0102] In the case where the selection for supplying of the rotational fluid Fa continuously was made earlier in Step **109**, the rotational fluid Fa is being supplied currently, and the cleaning object S is being rotated by the effect of the rotational fluid Fa. Therefore, a negative determination NO is made in Step **115**, and the control may proceed directly to Step **117** without detouring to Step **116**.

[0103] Similarly to the cleaning sequence described earlier, in the drying sequence, the cleaning object S is rotated by the rotational fluid Fa supplied from the supporting units **11**. Therefore, with a small number of mechanically operable parts, the cleaning apparatus **10** may be provided in a compact and simple structure.

[0104] In Step **117**, the cleaning object S having been cleaned is dried. The cleaning object S may be dried in Step **117** with use of the cleaning unit **12**. As shown in FIG. **3**, the cleaning unit **12** jets the air supplied from the air supply source **28** downward through the cleaning nozzle **26** and dries the cleaning object S by the jetted air. In the exemplary configuration in FIG. **3**, the second surface Sb of the cleaning object S may be the main target to be dried by the cleaning unit **12**. Meanwhile, centrifugal force by the rotation of the cleaning object S may remove the liquid droplets from not only the second surface Sb but also the first surface Sa and the side surface Sc; therefore, the cleaning object S may be dried entirely.

[0105] By the cleaning nozzle **26** being movable to the position above the outer edge of the cleaning object S, the air from the cleaning nozzle **26** may be delivered not only to the second surface Sb but also to the side surface Sc of the cleaning object S and a part of the first surface Sa continuous from the side surface Sc to dry.

[0106] The cleaning unit **12** jets the air downward from the cleaning nozzle **26** located above the cleaning object S to apply moderate pressure from above to the cleaning object S, thereby preventing the cleaning object S, which may otherwise deviate upward from the supporting units **11**, from being elevated excessively. Moreover, it is the air from the cleaning nozzle **26** that is supplied by the cleaning unit **12**; therefore, the cleaning object S may not be pressed excessively intensely against the supporting surfaces **16** from above.

[0107] The cleaning object S may be dried using the rotational fluid Fa as the drying gas in the drying process in Step **117**. In this case, as a first selection, for example, the air jetted from the cleaning nozzle **26** and the rotational fluid Fa jetted from the first supply ports **20** may be used in combination for drying. For another example, as a second selection, without supplying the air from the cleaning nozzle **26**, the rotational fluid Fa supplied from the first supply ports **20** may be jetted as the air to dry the cleaning object S. In the case of the second selection, the cleaning unit **12** (the cleaning nozzle **26**, the cleaning-water supply source **27**, and the air supply source **28**) may be omitted, and the supporting units **11** may serve to provide the dryer function of the cleaning unit (the configuration to jet the rotational fluid Fa may serve as the cleaning unit).

[0108] In either the first selection or the second selection, the floater fluid Fb in addition to the rotational fluid Fa may be used as the gas for drying, optionally. By using both the rotational fluid Fa and the floater fluid Fb as the dryer gas, the drying effect may be improved even more.

[0109] As such, how the drying gas may be supplied in the drying process in Step **117** may be selected from multiple options. The rotational fluid Fa has been from the first supply ports **20** continuously from Step **107** or Step **116**; therefore, in Step **117**, the controller may merely need to select whether the dryer air should be supplied from the cleaning nozzle **26** or not, in the case where the cleaning apparatus **10** is equipped with the cleaning nozzle **26**. Therefore, a complicated control is not required.

[0110] The drying process in Step **117** may be terminated on a predetermined time basis or may be ended based on a predetermined condition other than time. The condition other than time may include, for example, a degree of dryness of the cleaning object, which may be determined based on information indicating dryness of the cleaning object S estimated from, for example, a reflection rate in a captured image of the cleaning object S, so that the controller may continue the drying process until the controller determines that the cleaning object S is substantially dry.

[0111] After the drying process in Step **117** is completed, the control proceeds to Step **118**. In Step **118**, supply of the rotational fluid Fa from the first supply ports **20** is discontinued. As supply of the rotational fluid Fa is discontinued, the cleaning object S stops rotating and stays to be supported on the supporting surfaces **16** of the supporting units **11**.

[0112] Next, the controller determines a supplying status of the floater fluid Fb in Step **119**. In a case where the drying process was performed without supplying the floater fluid Fb to the cleaning object S in Step **117**, a negative determination NO is made in Step **119**, and the control proceeds to Step **120**. In Step **120**, the cleaning object S supported by the supporting units **11** is picked up by the conveyer assembly and removed from the cleaning apparatus **10** by the conveyer assembly.

[0113] In a case where the drying process was performed with the floater fluid Fb being supplied to the cleaning object S to elevate the cleaning object S in Step **117**, a positive determination YES is made in Step **119**, and the control proceeds to Step **121**. In Step **121**, the controller determines a setting of timing when to remove the cleaning object S from the supporting units **11**.

[0114] In a case where the cleaning object S is removed from the supporting units **11** after supply of the floater fluid Fb is discontinued (NO in Step **121**), the control proceeds to Step **122**, where supply of the floater fluid Fb from the second supply ports **23** is discontinued, and thereafter, in Step **123**, the cleaning object S is removed by the conveyer assembly from the supporting units **11**.

[0115] In a case where the cleaning object S is removed from the supporting units **11** before supply of the floater fluid Fb is discontinued (YES in Step **121**), the control proceeds to Step **124**, where the cleaning object S is removed by the conveyer assembly from the supporting units **11**, and thereafter, in Step **125**, supply of the floater fluid Fb from the second supply ports **23** is discontinued.

[0116] The supporting units **11** are configured to support the cleaning object S without covering the second surface Sb; therefore, in Step **123** or Step **124**, the conveyer assembly may pick up and hold the cleaning object S by the second surface Sb to remove from the supporting units **11** regardless of whether the cleaning object S is elevated by the floater fluid Fb or not.

[0117] The drying sequence is completed by performing one of Step **120**, **123**, or **125**. Once the drying sequence is completed, the control exits the flowchart shown in FIG. **6**, and the series of operations in the cleaning apparatus **10** ends thereat.

[0118] As described above, according to the cleaning apparatus **10** of the first embodiment and the cleaning method using the cleaning apparatus **10**, the cleaning object S is rotated by the rotational fluid Fa supplied from the first supply ports **20**. Therefore, a complex mechanical structure, such as a supporting mechanism using a shaft to rotate the cleaning object S or a transmission mechanism to transmit power of a motor, is not required, and the cleaning apparatus **10** may be provided in a downsized and simple structure to clean the cleaning object S efficiently.

[0119] Moreover, the cleaning apparatus **10** may elevate the cleaning object S by the floater fluid Fb supplied from the second supply ports **23** so that the cleaning object S may be cleaned in the state not contacting the supporting units **11**. Therefore, the cleaning object S may be rotated at a high speed by a low load. Moreover, the cleaning object S may be prevented from being dirtied or damaged by contact with the supporting units **11**.

[0120] In particular, by starting to supply the floater fluid Fb to the supporting surfaces **16** of the supporting unit **11** before the cleaning object S is placed thereon, and by discontinuing the supply of the floater fluid Fb after the cleaning object S is separated from the supporting surfaces **16** of the supporting units **11** to be removed from the supporting units **11**, the series of processes may be completed while the cleaning object S remains contactless throughout the processes.

[0121] The rotational fluid Fa and/or the floater fluid Fb may be used as complements to the functions of the cleaning water and for the dryer air to be supplied from the cleaning unit **12**, thereby improving the efficiency of cleaning and drying. Moreover, optionally, the cleaning apparatus **10** may be configured such that the rotational fluid Fa or the floater fluid Fb may be used as the cleaning water or the dryer air (applying a structure to supply the rotational fluid Fa or the floater fluid Fb to function as the cleaning unit), without being equipped with the configuration of the cleaning unit **12**. In the case where the cleaning object S is cleaned with the rotational fluid Fa, the cleaning unit may at least include the first supply ports **20** for supplying the rotational fluid Fa.

[0122] In the above description, the cleaning sequence in FIG. **5** and the drying sequence in FIG. **6** are performed consecutively; however, optionally, the cleaning sequence alone may be performed in the cleaning apparatus **10**, and the drying sequence may be performed in an apparatus different from the cleaning apparatus **10**, or the cleaning object S may be dried naturally.

[0123] FIGS. **7** and **8** show a cleaning apparatus **30** according to a second embodiment. The cleaning apparatus **30** includes, further to the configuration of the cleaning apparatus **10** in the first embodiment, lids **31** arranged to face the outer edge on the second surface Sb of the cleaning object S. The configuration of the cleaning apparatus **30** is the same as that of the cleaning apparatus **10** except for the configuration of the lids **31**; therefore, description of the configuration common to the cleaning apparatus **10** is herein omitted.

[0124] The lids **31** are in a form of a ring divided by the communicating portions **13** into two along the circumferential direction, similarly to the supporting units **11**. The lids **31** each include an inner ring portion **32** located on an inner circumferential side and an outer ring portion **33** located on an outer circumferential side. The inner ring portion **32** and the outer ring portion **33** are formed integrally.

[0125] The outer ring portion **33** is supported on top of the side portion **15** of the supporting unit **11**. The inner ring portion **32** has a greater thickness in the vertical direction than the outer ring portion **33** and is fitted to the step shape in the supporting unit **11** formed between the supporting surface **16** of the supporting portion **14** and the inner circumferential surface **18** of the side portion **15**. As the lids **31** are attached from above to the supporting units **11** supporting the cleaning object S, as shown in FIG. **8**, the inner ring portions **32** are located above the outer edge (area on the second surface Sb closer to the side surface Sc) of the cleaning object S. With the outer ring portions **33** being placed on the upper surfaces of the side portions **15**, the inner ring portions **32** face the supporting portions **14** in the vertical direction with a predetermined amount of gap maintained there-between. As such, the inner ring portions **32** is prevented from contacting the cleaning object S, so that the rotation of the cleaning object S may not be hindered, or from covering the first supply ports **20**, so that the jetting of the rotational fluid Fa may not be hindered.

[0126] In the inner ring portion **32** in each lid **31**, a plurality of third supply ports **34** are formed to open downward. The plurality of third supply ports **34** are arranged at predetermined intervals in the circumferential direction of the lid **31** (in the rotating direction of the cleaning object S) at positions above the second supply ports **23** formed in the supporting portion **14** in the supporting unit **11**. The plurality of third supply ports **34** in this arrangement are located to be open toward the second surface Sb of the cleaning object S at positions in the vicinity of the outer edge of the cleaning object S supported by the supporting units **11**. The number of the plurality of second supply ports **23** and circumferential intervals thereof coincide with the number of the third supply ports **34** and the circumferential intervals thereof, and as shown in FIG. **8**, the second supply ports **23** and the third supply ports **34** are respectively arranged coaxially and aligned vertically.

[0127] Inside each lid **31**, a fluid supply path **35** extending in the circumferential direction is formed. The plurality of third supply ports **34** are formed to branch off from the fluid supply path **35** and extend downward. The third supply ports **34** each extend substantially vertically.

[0128] The cleaning apparatus **30** is provided with a fluid supply source **36** to supply the fluid to the fluid supply paths **35** and is connected to the fluid supply paths **35** via an open/close valve. The fluid to be supplied by the fluid supply source **36** is at least one of a liquid or a gas, preferably water, when the fluid is a liquid, or air, when the fluid is a gas. Optionally, depending on the material or the type of the cleaning object S, a fluid other than water or air may supplied from the fluid supply source **36**.

[0129] In a case where the fluid supply source **36** is capable of supplying both a liquid and a gas, the fluid supply source **36** may be configured to have a liquid supply source and a gas supply source, and the type of the fluid to be supplied to the fluid supply paths **35** may be selected by an operation of an open/close valve in a flow path connecting to the liquid supply source and the gas supply source. Optionally, a mixed fluid (two fluids) containing both a liquid and a gas may be supplied from the fluid supply source **36**.

[0130] On one of end faces of each lid **31** in the circumferential direction facing the communicating portion **13**, an inlet **351**, from which the fluid supplied from the fluid supply source **36** may be drawn into the fluid supply path **35**, is formed, and on the other of the end faces of each lid **31** in the circumferential direction facing the communicating portion **13**, an outlet **352**, from which the fluid flowed through the fluid supply path **35** may be discharged, is formed.

[0131] The fluid supplied from the fluid supply source **36** to the inlet **351** of each fluid supply path **35** may travel inside the fluid supply path **35** in an arc shape in a plan view. The fluid flowing through the fluid supply paths **35** may enter the individual third supply ports **34** and may be jetted out from outlets formed on the lower surface of the inner ring portions **32**.

[0132] The fluid jetted from the third supply port **34** will be herein called a hold-down fluid Fc. In FIG. **8**, the hold-down fluid Fc is represented schematically in an arrow. The hold-down fluid Fc jetted from the third supply ports **34**, which extend in the vertical direction, flows downward and hits the second surface Sb of the cleaning object S. When cleaning, as the cleaning object S is elevated by the floater fluid Fb jetted from the second supply ports **23**, the hold-down fluid Fc jetted from the third supply ports **34** may hold down the cleaning object S from above, restricting the cleaning object S from floating excessively from the supporting portions **14** of the supporting units **11**. As such, the cleaning object S may be held stably at a constant position in the vertical direction and rotated by the balance between the forces of the floater fluid Fb and the hold-down fluid Fc.

[0133] The flowing speed and the flowing rate of the floater fluid Fb are set depending on the pressure or the amount of the fluid to be supplied from the fluid supply source **25**, and the flowing speed and the flowing rate of the hold-down fluid Fc are set depending on the pressure or the amount of the fluid to be supplied from the fluid supply source **36**. Therefore, by the controller controlling the cleaning apparatus **30** and the behaviors of the fluid supply source **25** and the fluid supply source **36**, the balance between the force of the floater fluid Fb to elevate the cleaning object S and the force of the hold-down fluid Fc to restrict the cleaning object S from floating may be adjusted, and the cleaning object S may be rotated in the elevated state between the supporting units **11** and the lids **31**.

[0134] In order to prevent the cleaning object S from flapping or bending, it is preferable that the second supply ports **23** and the third supply ports **34** are located coaxially aligning in the vertical direction, as shown in FIG. **8**, so that the positions where the floater fluid Fb hits the cleaning object S and the positions where the hold-down fluid Fc hits the cleaning object S are aligned. However, as long as the cleaning object S is elevated and rotated preferably between the supporting units **11** and the lids **31**, the arrangement is not necessarily limited as above, or the positions of the second supply ports **23** and the third supply ports **34** in the circumferential direction may be offset from each other.

[0135] The inner ring portion **32** in each lid **31** has a discharging portion **37** in a form of a step located at a position higher than the positions of the openings of the third supply ports **34**, on the inner circumferential side with respect to the area where the third supply ports **34** are formed. In other words, at a position on the inner circumferential side and adjacent to the third supply ports **34**, the discharging portions **37** to form a larger gap with the cleaning object S than the third supply ports **34** are provided. With the lids **31** attached to the supporting units **11**, the discharging portions **37** of the lids **31** are located to face toward the discharging portions **17** of the supporting units **11** in the vertical direction. Note that the discharging portions **37** are not required but are optional.

[0136] With this arrangement, the rotational fluid Fa, the floater fluid Fb, and the hold-down fluid Fc supplied to the areas in the vicinity of the outer edge of the cleaning object S may be easily discharged toward the area on the inner circumferential side of the lids **31** (the opening portion **19** in the supporting units **11**) through the discharging portions **37**. As such, the rotational fluid Fa, the floater fluid Fb, and the hold-down fluid Fc may be prevented from staying in the areas in the vicinity of the outer edge of the cleaning object S, and the cleaning object S may be stabilized in the posture and rotated accurately.

[0137] Cleaning operations and a method performed with the cleaning apparatus **30** configured as above are the same as those of the cleaning apparatus **10** described above except for the details concerning the lids **31**; therefore, differences from the cleaning apparatus **10** will be briefly explained below with reference to the flowcharts shown in FIGS. **5** and **6**.

[0138] After the cleaning object S is conveyed to the supporting units **11** and placed on the supporting surfaces **16** in the supporting step in one of Step **101**, Step **104**, or Step **106**, the lids **31** are attached on top of the supporting units **11**. The lids **31** are placed in a state such that the outer ring portions **33** are placed and supported on the upper surfaces of the side portions **15**, and the inner ring portions **32** face the second surface Sb of the cleaning object S with a gap maintained there-between. Moreover, the lids **31** may be fixed to the supporting units **11** with fixing members (such as bolts and nuts or clamps), which are not shown.

[0139] In the cleaning sequence, in a case where the selection not to supply the floater fluid Fb from the second supply ports **23** is made in Step **100** (NO in Step **100**), it is preferable that the hold-down fluid Fc may not be supplied to the cleaning object S from the third supply ports **34**. Thereby, the cleaning object S may be prevented from being urged intensely against the supporting surfaces **16**.

[0140] In a case where the selection to supply the floater fluid Fb is made in Step **100** (YES in Step **100**), an option of supplying the hold-down fluid Fc from the third supply ports **34** or an option of not supplying the hold-down fluid Fc from the third supply ports **34** is selectable.

[0141] By supplying the hold-down fluid Fc from the third supply ports **34** to the cleaning object S in the cleaning sequence, the floater fluid Fb may elevate the cleaning object S while the hold-down fluid Fc holds the cleaning object S down so that the cleaning object S may be held at a constant height position and rotated stably.

[0142] In the case where the hold-down fluid Fc is not supplied from the third supply ports **34** in the cleaning sequence, the lids **31** may serve as physical lids, by which the cleaning object S may be prevented from deviating largely upward from the supporting units **11**.

[0143] In the case where the hold-down fluid Fc is supplied from the third supply ports **34** in the cleaning sequence, in order to prevent the cleaning object S from adherence of foreign matter thereto due to dryness, it is preferable that the hold-down fluid Fc is a liquid (or two fluids). However, in a case where a gas is used as the floater fluid Fb in the cleaning sequence, use of a gas as the hold-down fluid Fc to match with the use of the floater fluid Fb being a gas is not necessarily excluded.

[0144] In Step **111** after cleaning in Step **108**, further to the selection whether supplying of the floater fluid Fb is to be continued or discontinued, a selection whether supplying of the hold-down fluid Fc is to be continued or discontinued is made. Conditions for whether to continue or discontinue supplying of the hold-down fluid Fc are the same as those in the case of the floater fluid Fb described above. When supplying of the hold-down fluid Fc is to be discontinued, the process to discontinue supplying of the hold-down fluid Fc is performed in the same manner as the process to discontinue supplying of the floater fluid Fb in Step **112**.

[0145] In the drying sequence, in a case where the selection not to supply the floater fluid Fb from the second supply ports **23** is made in Step **113** (NO in Step **113**), it is preferable that the hold-down fluid Fc may not be supplied to the cleaning object S from the third supply ports **34**. Thereby, the cleaning object S may be prevented from being urged intensely against the supporting surfaces **16**.

[0146] In a case where the selection to supply the floater fluid Fb is made in Step **113** (YES in Step **113**), an option of supplying the hold-down fluid Fc from the third supply ports **34** or an option of not supplying the hold-down fluid Fc from the third supply ports **34** is selectable.

[0147] By supplying the hold-down fluid Fc from the third supply ports **34** to the cleaning object S in the drying sequence, the floater fluid Fb may elevate the cleaning object S while the hold-down fluid Fc holds the cleaning object S down so that the cleaning object S may be held at a constant height position and rotated stably.

[0148] In the case where the hold-down fluid Fc is not supplied from the third supply ports **34** in the drying sequence, the lids **31** may serve as physical lids, by which the cleaning object S may be prevented from deviating largely upward from the supporting units **11**.

[0149] In the case where the hold-down fluid Fc is supplied in the drying sequence, a gas, rather than liquid that may wet the cleaning object S, is selected as the hold-down fluid Fc.

[0150] After the drying process in Step **117** is completed and, in Step **118**, supply of the rotational fluid Fa from the first supply ports **20** is discontinued, and after the cleaning object S stops rotating, the lids **31** are removed from the supporting units **11**. In a case where the hold-down fluid Fc was being supplied continuously in the drying sequence, the lids **31** are removed from the supporting units **11** after supply of the hold-down fluid Fc is discontinued. By removing the lids **31**, the inner ring portions **32** of the lids **31** no longer cover the cleaning object S from above. Therefore, the cleaning object S may be removed from the supporting units **11** by performing Step **119** and subsequent processes after the lids **31** are removed.

[0151] As described above, according to the cleaning operation and the cleaning method using the cleaning apparatus **30** of the second embodiment, which is provided with the lids **31** to cover the cleaning object S from above and is configured to hold the cleaning object S down by the hold-down fluid Fc jetted from the third supply ports **34** in the lids **31**, the cleaning object S may be cleaned and dried while being securely supported by the supporting units **11** without deviating therefrom.

[0152] FIGS. **9** and **10** show a cleaning apparatus **40** according to a third embodiment. The cleaning apparatus **40** includes, further to the configuration of the cleaning apparatus **30** in the second embodiment, a second cleaning unit **41** configured to clean the first surface Sa of the cleaning object S. In other words, the cleaning apparatus **40** is configured to clean both the first surface Sa and the second surface Sb simultaneously by the cleaning unit **12** and the second cleaning unit **41**, which are arranged above and below the cleaning object S, respectively. The configuration of the cleaning apparatus **40** is the same as that of the cleaning apparatus **30** except for the configuration of the second cleaning unit **41**; therefore, description of the configuration common to the cleaning apparatus **30** is herein omitted. Note that in FIG. **10**, the cleaning apparatus **40** is illustrated with the lids **31** being removed, and the illustration of the lids **31** is omitted.

[0153] The second cleaning unit **41** has a contact-cleaning device **42** and a cleaning nozzle **43** that are located below the cleaning object S being supported by the supporting units **11**. The contact-cleaning device **42** includes a cylindrical body formed of, for example, a sponge or a brush and is supported rotatably around a horizontally extending supporting shaft **44**. As shown in FIG. **10**, the contact-cleaning device **42** and the supporting shaft **44** are arranged through the opening portion **19** in the supporting units **11**, with both ends of the contact-cleaning device **42** being located in the communicating portions **13**.

[0154] A cleaning-water supply source **45** and an air supply source **46** are connected to the cleaning nozzle **43** via respective open/close valves. The cleaning nozzle **43** may jet cleaning water supplied

from the cleaning-water supply source **45**, air supplied from the air supply source **46**, or a mixed fluid containing cleaning water and air (two fluids) upward.
[0155] The contact-cleaning device **42** and cleaning nozzle **43** in the second cleaning unit **41** are cleaning devices that may be located at positions to face the first surface Sa of the cleaning object S which is supported by the supporting units **11**. For cleaning the cleaning object S, cleaning water may be jetted from the cleaning nozzle **43** at the cleaning object S, and the contact-cleaning device **42** may contact the cleaning object S, thereby the first surface Sa of the cleaning object S may be cleaned. Optionally, the second cleaning unit **41** may be configured such that the contact-cleaning device **42** is rotatable passively by a frictional force generated between the contact-cleaning device **42** and the cleaning object S rotated by the rotational fluid Fa or such that the contact-cleaning device **42** is driven to rotate by a driving force transmitted to the supporting shaft **44** from, for example, a motor.
[0156] The contact-cleaning device **42** may contact the first surface Sa of the cleaning object S with appropriate contact pressure that may not cause an excessive friction force so as not to impede smooth rotation of the cleaning object S by the rotational fluid Fa. In order to adjust the contact pressure of the contact-cleaning device **42** against the cleaning object S, the supporting shaft **44** may be configured to be movable in the vertical direction, and the controller may preferably change the vertical position of the contact-cleaning device **42**.
[0157] As shown in FIG. **10**, a length of the contact-cleaning device **42** in an axial direction is greater than the diameter of the cleaning object S, and the contact-cleaning apparatus **42** is located to traverse the cleaning object S through the center in a direction of the diameter of the cleaning object S in a plan view. As such, by rotating the cleaning object S relatively to the supporting units **11**, a cleaning range of the contact-cleaning apparatus **42** may entirely cover the first surface Sa.
[0158] The ends of the contact-cleaning apparatus **42** are both located in the spaces of the communicating portions **13**. This allows the contact-cleaning apparatus **42**, which is longer than the diameter of the cleaning object S and capable of cleaning the entire surface of the first surface Sa, to be arranged in the supporting units **11** without interfering therewith. As such, by using the spaces of the communicating portions **13**, which are for discharging the fluid outward from the opening portion **19** in the supporting units **11**, as spaces to accommodate the ends of the contact-cleaning apparatus **42**, a structure with preferable space efficiency is provided. As the contact-cleaning apparatus **42** is located to be lower than the lids **31**, the lids **31** may not interfere with the contact-cleaning apparatus **42**, even if, for example, the lids **31** are in a form not divided by the communicating portions **13**. In other words, in the light of prevention of interference with the contact-cleaning apparatus **42**, it is sufficient that at least the supporting units **11** have the communicating portions **13**.
[0159] FIG. **11** shows a cleaning apparatus **50** according to a fourth embodiment. The cleaning apparatus **50** is based on the configuration of the cleaning apparatus **30** according to the second embodiment but is different in that the lids **31** are openable/closable with respect to the supporting units **11** via rotating assemblies **51**. The lids **31** are in a configuration, as those in FIG. **7**, divided in the circumferential direction by the communication portions **13** into two parts. Each of the lids **31** is pivotable to be opened or closed with respect to the supporting unit **11** via the respective rotating assembly **51**.
[0160] The rotating assemblies **51** each support the lid **31** pivotably about an axis extending in the horizontal direction, and the lids **31** are pivoted by a driving force of, for example, a rotary cylinder or a motor. The lids **31**, when at a closed position, cover the outer edge of the cleaning object S supported by the supporting units **11**.
[0161] As the rotating assemblies **51** are driven to open the lids **31** upward, as shown in FIG. **11**, the lids **31** uncover the areas above the outer edge of the cleaning object S supported by the supporting units **11**, and the supporting units **11** may be loaded or unloaded with the cleaning object S using the conveyer assembly including, for example, a conveyer arm **52** that may suction and

hold the second surface Sb of the cleaning object S. The conveyer arm **52** may be of a type that jets air from a holder surface (lower surface) and suctions to hold the cleaning object S in a contactless manner using the suctioning effect according to Bernoulli's theorem, or of a type that suctions to hold the cleaning object S by suctioning the air from suction holes formed in the holder surface (lower surface). By using the contactless-typed conveyer arm **52** and supporting the cleaning object S in the contactless manner using the floater fluid Fb in the cleaning apparatus **50**, the cleaning object S may be handled in every stage in the process including loading and unloading without mechanically contacting the cleaning object S.

[0162] When the cleaning apparatus **50** is loaded or unloaded with the cleaning object S, the cleaning unit **12** is operated to withdraw the cleaning nozzle **26** from the position above the opening portion **19** in the supporting units **11** so that cleaning nozzle **26** may not interfere with the conveyer arm **52**. FIG. **11** shows the cleaning nozzle **26** being withdrawn.

[0163] According to the cleaning apparatus **50**, it is not necessary to attach or detach the lids **31** each time the cleaning apparatus **50** is loaded or unloaded with the cleaning object S. Therefore, while taking the advantage of the lids **31** when the cleaning object S is being cleaned, the process may be speeded up in the preparatory stage for cleaning and the post-cleaning stage so that the operation of the apparatus may be streamlined.

[0164] In the embodiments described above, the lids **31** are in the form divided in the circumferential direction into two pieces. However, optionally, the lids **31** may be configured as a lid **31** formed of a not divided single piece pivotably supported by a single rotating assembly **51** or as lids **31** divided in the circumferential direction into three or more pieces pivotably supported by three or more rotating assemblies **51**.

[0165] FIGS. **12** and **13** show a cleaning apparatus **60** according to a fifth embodiment. In the first through fourth embodiments, the cleaning object S is rotated by the rotational fluid Fa jetted from the first supply ports **20** formed on the inner circumferential surfaces **18** of the side portions **15** of the supporting units **11**. In contrast, in the cleaning apparatus **60**, the cleaning object S is rotated by a rotational fluid Fd jetted from a plurality of first supply ports **61**, which are formed in the supporting portions **14** of the supporting units **11** and are open on the supporting surfaces **16**.

[0166] The plurality of first supply ports **61** are arranged at predetermined intervals in the circumferential direction of the supporting units **11** (in the rotating direction of the cleaning object S), and the outlets of the first supply ports **61** are located in the vicinity of the outer edge of the cleaning object S supported by the supporting units **11** and are open toward the first surface Sa of the cleaning object S.

[0167] Inside the supporting portion **14** in each supporting unit **11**, a fluid supply path **62** extending in the circumferential direction is formed. The fluid supply paths **62** forms an annular flow path located on the inner side in the circumferential direction with respect to the inner circumferential surface **18** of the side portion **15**, having an arc form. A plurality of first supply ports **61**, which are flow paths branching off from the fluid supply path **62** and extend obliquely upward to the supporting surface **16**, are formed at predetermined intervals in the circumferential direction.

[0168] The cleaning apparatus **60** is provided with a fluid supply source **63** to supply the fluid to the fluid supply paths **62** and is connected to the fluid supply paths **64** via an open/close valve. The fluid supply source **63** includes a tank to store the fluid and a pump to pump the fluid from the tank. The fluid to be supplied by the fluid supply source **63** is at least one of a liquid or a gas, preferably water, when the fluid is a liquid, or air, when the fluid is a gas. Optionally, depending on the material or the type of the cleaning object S, a fluid other than water or air may supplied from the fluid supply source **63**.

[0169] On one of the end faces of each supporting unit **11** in the circumferential direction facing the communicating portion **13**, an inlet **621**, from which the fluid supplied from the fluid supply source **63** may be drawn into the fluid supply path **62**, is formed, and on the other of the end faces of each supporting unit **11** in the circumferential direction facing the communicating portion **13**, an outlet **622**, from which the fluid flowed through the fluid supply path **62** may be discharged, is formed.

[0170] The fluid supplied from the fluid supply source **63** to the inlet **621** of each fluid supply path **62** may travel inside the fluid supply path **62** in the arc shape in the plan view. The fluid flowing through the fluid supply paths **62** may enter the individual first supply ports **61** and may be jetted out from outlets of the first supply ports **61** formed on the supporting surfaces **16**.

[0171] The fluid jetted from the first supply ports **61** will be herein called a rotational fluid Fd. As shown in FIG. **13**, each of the plurality of first supply ports **61** is a flow path branching off from the fluid supply path **62** and extending obliquely upward to the supporting surface **16**. In other words, each of the first supply ports **61** includes a flow path in a shape which includes a component (not parallel to the thickness direction of the cleaning object S) of the rotating direction of the cleaning object S supported by the supporting unit **11**. The rotational fluid Fd jetted along the shape of each first supply port **61** is oriented in a direction including the component of the rotating direction of the cleaning object S and may contact the first surface Sa from an obliquely lower position. Therefore, the rotational fluid Fd may hit the first surface Sa in the vicinity of the outer edge of the cleaning object S so that the force in the rotating direction acts on the cleaning object S, and the cleaning object S supported by the supporting units **11** may be rotated by the force of the rotational fluid Fd.

[0172] When the amount and the force of the fluid to be supplied are equal, the configurations of the cleaning apparatuses **10**, **30**, **40**, **50** of the first through fourth embodiments, which use the rotational fluid Fa jetted from the first supply ports **20** located on the exterior of the side surface Sc of the cleaning object S, may transmit the force in the rotating direction to cleaning object S more efficiently than the configuration of the cleaning apparatus **60** according to the fifth embodiment, which uses the rotational fluid Fd jetted from the first supply ports **61** toward the first surface Sa of the cleaning object S. Therefore, in the cleaning apparatus **60**, the amount and the speed of the flow and/or the type of the rotational fluid Fd are adjusted in consideration of the positions of the first supply ports **61** so that the rotating force in sufficient intensity may be applied to the cleaning object S.

[0173] For example, by using a liquid having a higher viscosity than a gas as the rotational fluid Fd, a substantial rotating force may be applied to the cleaning object S. Moreover, the rotational fluid Fd being a liquid may have the effect of elevating the cleaning object S from the supporting surfaces **16** so that the rotational fluid Fd may work also as the floater fluid Fb.

[0174] On the other hand, in a case where a gas is used as the rotational fluid Fd, if the speed of the rotational fluid Fd flowing along the first surface Sa is high, a negative pressure may be generated between the supporting surfaces **16** and the first surface Sa by Bernoulli's theorem, and the cleaning object S may be rotated while being attracted to (but without contacting) the supporting surfaces **16**. In this case, floating of the cleaning object S may be regulated without using the hold-down fluid Fc described earlier.

[0175] Unlike the cleaning apparatus **60** of the present embodiment, optionally, the cleaning object S may be rotated by jetting a rotational fluid toward the second surface Sb rather than the first surface Sa of the cleaning object S. For example, the lids **31** described above may be added to the cleaning apparatus **50**, and the first supply ports **61** to replace the third supply ports **34** may be formed in the lids **31**. Thereby, the rotational fluid that may work similarly to the rotational fluid Fd may be jetted at the second surface Sb.

[0176] As shown in FIG. **12**, the rotational fluid Fd jetted from the first supply ports **61** contains not only a component to flow in the rotating direction of the cleaning object S but also a component to proceed toward the outer circumferential side of the cleaning object S in a plan view. In other words, the first supply ports **61** are each in a form with inclination, so as to expand toward the outer circumferential side of the supporting units **11**, with respect to a tangent to the flowing direction Ta of the fluid flowing in the fluid supply path **62**.

[0177] When the rotational fluid Fd is jetted at the first surface Sa of the cleaning object S, and if the fluid (in particular, air) concentrates at a center on the first surface Sa, the flow may collide with air currents in the surrounding area and stagnate thereat, causing positive pressure in the area below the center of the first surface Sa. As the positive pressure grows in the area below the center of the first surface Sa, the cleaning object S may rapidly move in the direction to separate from the supporting surfaces **16** and may deviate from the state being supported by the supporting units **11**.
[0178] In contrast, the structure of the first supply ports **61** described above may release the rotational fluid Fd toward the outer circumferential side of the cleaning object S effectively, preventing the rotational fluid Fd jetted from the first supply ports **61** from concentrating at the center of the cleaning object S, thereby enabling the cleaning object S to stay supported by the supporting units **11** stably.
[0179] The cleaning apparatus **60** shown in the drawings illustrates a case where the cleaning object S is rotated by the rotational fluid Fd supplied from the first supply ports **61** alone in the supporting portions **14**. Optionally, however, the first supply ports **20** may be formed in the side portions **15** to supply the rotational fluid Fa described above, and the cleaning object S may be rotated by the rotational fluid Fd in combination with the rotational fluid Fa.
[0180] In the case where the cleaning object S is rotated by the rotational fluid Fd supplied from the first supply ports **61** alone, optionally, a configuration of the supporting units **11** with no side portions **15** may be applicable. In this case, however, the position of the cleaning object S in the horizontal direction may not be stabilized by the supporting portions **14** alone. Therefore, in order to prevent the cleaning object S from deviating in the horizontal direction, it is preferable that the side portions **15** are provided.
[0181] Optionally, the rotational fluid Fd may be used for cleaning the cleaning object S. In this arrangement, the cleaning unit to clean the cleaning object S incudes at least the first supply ports **61** for supplying the rotational fluid Fd.
[0182] FIGS. **14** and **15** show a cleaning apparatus **70** according to a sixth embodiment. The cleaning apparatus **70** includes a disk-shaped central supporting unit **71** in the opening portion **19** of the supporting units **11**. The central supporting unit **71** has a supporting surface **72** facing upward, on which the central portion of the first surface Sa of the cleaning object S may be supported. A height of the supporting surface **72** of the central supporting unit **71** is set at a height position, which is the height same as or higher than the supporting surfaces **16** of the supporting units **11**.
[0183] In the cleaning apparatus **70**, a rotational fluid Fe may be jetted at the cleaning object S from a plurality of first supply ports **73**, which are open on the supporting surface **72** of the central supporting unit **71**, to rotate the cleaning object S. The plurality of first supply ports **73** are formed at predetermined intervals in a circumferential direction (in the rotating direction of the cleaning object S) of the central supporting unit **71**, and outlets of the first supply port **73** are located to be open toward the first surface Sa of the cleaning object S supported on the central supporting unit **71**.
[0184] Inside the central supporting unit **71**, a fluid supply path **74** extending in the circumferential direction is formed. The plurality of first supply ports **73**, which are flow paths branching off from the fluid supply path **74** and extending obliquely upward toward the supporting surface **72**, are formed at predetermined intervals in the circumferential direction. Note that the central supporting unit **71** is not in a ring shape; therefore, the fluid supply path **74** extending therein is not necessarily in a form of an arc extending in the circumferential direction of the central supporting unit **71**.
[0185] The cleaning apparatus **70** is provided with a fluid supply source **75** to supply the fluid to the fluid supply path **74** and is connected to the fluid supply path **74** via an open/close valve. The fluid supply source **75** includes a tank to store the fluid and a pump to pump the fluid from the tank. The fluid to be supplied by the fluid supply source **75** is at least one of a liquid or a gas.
[0186] The fluid supplied from the fluid supply source **75** to the fluid supply path **74** may travel

inside the fluid supply path **74** in the arc shape in a plan view in the flowing direction Ta (see FIG. **15**). The fluid flowing through the fluid supply path **74** may enter the individual first supply ports **73** and may be jetted out from the outlets of the first supply ports **73** formed on the supporting surface **72** of the central supporting unit **71**.
[0187] The fluid jetted from the first supply ports **73** will be herein called a rotational fluid Fe. Similarly to the first supply ports **61** (see FIG. **13**) in the fifth embodiment, each of the plurality of first supply ports **73** is a flow path branching off from the fluid supply path **74** and extending obliquely upward to the supporting surface **72**. In other words, the first supply ports **73** has a flow path in a shape including a component (not parallel to the thickness direction of the cleaning object S) of the rotating direction of the cleaning object S supported by the central supporting unit **71**. The rotational fluid Fe jetted along the shape of each first supply port **73** is oriented in a direction including the component of the rotating direction of the cleaning object S and may contact the first surface Sa from an obliquely lower position. Therefore, the force in the rotating direction acts on the cleaning object S, and the cleaning object S supported by the central supporting unit **71** may be rotated by the force of the rotational fluid Fe.
[0188] Optionally, the rotational fluid Fe may be used for cleaning the cleaning object S. In this arrangement, the cleaning unit to clean the cleaning object S incudes at least the first supply ports **73** for supplying the rotational fluid Fe.
[0189] By using a liquid having a higher viscosity than a gas as the rotational fluid Fe, the effect to elevate the cleaning object S from the supporting surface **72** is achieved from the rotational fluid Fe while the rotational fluid Fe rotates the cleaning object S. Moreover, using a gas with a higher flow speed flowing along the first surface Sa as the rotational fluid Fe, by Bernoulli's theorem, the cleaning object S may be rotated while being attracted to the supporting surface **72**.
[0190] According to the configuration in which the rotational fluid Fe is jetted from the central supporting unit **71**, the rotational fluid Fe may be supplied to the cleaning object S without considering the shape of the outer edge of the cleaning object S for applying the rotating force to the cleaning object S; therefore, a cleaning object having a shape other than the disk shape such as the illustrated cleaning object S may be selected to be used in the cleaning apparatus **70**. For example, a rectangular package substrate may be used as the cleaning object to be rotated.
[0191] As shown in FIG. **15**, the first supply ports **73** are each in a form with inclination, so as to expand toward the outer circumferential side of the central supporting unit **71**, with respect to a tangent to the flowing direction Ta of the fluid flowing through the fluid supply path **74**. As such, the rotational fluid Fe jetted from the first supply ports **73** contains not only a component to flow in the rotating direction of the cleaning object S but also a component to proceed toward the outer circumferential side of the cleaning object S in a plan view. This structure of the first supply ports **73** may release the rotational fluid Fe toward the outer circumferential side of the cleaning object S effectively, preventing the rotational fluid Fe jetted from the first supply ports **73** from concentrating at the center of the cleaning object S, thereby enabling the cleaning object S to stay supported by the central supporting unit **71** stably.
[0192] According to the cleaning apparatus **70** shown in FIGS. **14** and **15**, the ring-shaped supporting units **11** are located on the outer side of the central supporting unit **71**. Further to the rotational fluid Fe to be supplied from the plurality of first supply ports **73** formed in the central supporting unit **71**, optionally, the rotational fluid Fa to be jetted from the first supply ports **20** formed in the supporting units **11** and/or the floater fluid Fb to be jetted from the plurality of second supply ports **23** may be used in combination for supporting and rotating the cleaning object S.
[0193] Moreover, unlike the cleaning apparatus **70** shown in FIGS. **14** and **15**, without providing the supporting units **11**, the disk-shaped central supporting unit **71** alone may be provided as the supporting structure in the cleaning apparatus. In this case, however, the position of the cleaning object S in the horizontal direction may not be stabilized by the central supporting unit **71** alone.

Therefore, in order to prevent the cleaning object S from deviating in the horizontal direction, it is preferable that a configuration equivalent to the side portions **15** are provided.
[0194] As described above, according to the cleaning apparatuses and the cleaning method in the above embodiments, the cleaning object may be rotated by supplying the fluid, thereby enabling cleaning of the cleaning object efficiently in a downsized structure without requiring a complex rotating mechanism.
[0195] The technique to rotate a cleaning object with use of a fluid may be applied to a processing operation other than cleaning as described in the above embodiments. For example, the technique may be applied to a spin-coating process, in which a liquid resin is supplied to a surface of a plate-shaped object, and the object is rotated to cause the liquid resin to spread on the surface thereof by the centrifugal force.
[0196] Embodiment of the present disclosure may not necessarily be limited to the configurations described above or in the modified examples but may be modified, substituted, or altered in various ways without departing from the spirit of the technical idea of the present disclosure. Furthermore, if the technical idea of the present disclosure may be realized in a different way due to technological progress or other derived technology, it may be implemented with use of the method. Therefore, the claims cover all embodiments that may be included within the scope of the technical idea of the present disclosure.
[0197] As explained above, according to the cleaning apparatuses and the cleaning method of the present disclosure, a cleaning object may be rotated and cleaned by a rotational fluid. Therefore, the cleaning object may be cleaned efficiently by a cleaning apparatus in a downsized structure, and a cost for introducing the cleaning apparatus and restrictions on an installation location may be reduced.

Claims

1. A cleaning apparatus including a cleaning unit configured to clean a cleaning object, comprising: a supporting portion located on a side of a first surface of the cleaning object, the supporting portion being configured to support the cleaning object; and at least one first supply port, from which a rotational fluid is supplied to the cleaning object supported by the supporting portion, wherein the cleaning apparatus is configured to clean the cleaning object while the cleaning object is rotated by the rotational fluid supplied from the at least one first supply port.
2. The cleaning apparatus according to claim 1, wherein the supporting portion is configured to support an outer edge of the cleaning object and includes a side portion configured to cover an exterior of the cleaning object supported by the supporting portion.
3. The cleaning apparatus according to claim 2, wherein the at least one first supply port is formed on an inner circumference of the side portion.
4. The cleaning apparatus according to claim 3, further comprising: at least one second supply port provided on a supporting surface of the supporting portion, the supporting surface being configured to support the cleaning object thereon, wherein the cleaning unit is configured to clean the cleaning object in a state where the cleaning object is elevated from the supporting surface by a floater fluid supplied from the at least one second supply port.
5. The cleaning apparatus according to claim 1, wherein the cleaning unit is configured to clean at least one of the first surface, a second surface opposite to the first surface, or a side surface, of the cleaning object.
6. The cleaning apparatus according to claim 1, wherein the cleaning unit at least includes the at least one first supply port and is configured to clean the cleaning object with the rotational fluid.
7. The cleaning apparatus according to claim 1, wherein the cleaning unit includes, separately from the at least one first supply port, a cleaning device locatable at a position to face at least one of the first surface of the cleaning object supported by the supporting portion or a second surface opposite

to the first surface of the cleaning object, and the cleaning unit is configured to clean the cleaning object using the cleaning device.

8. The cleaning apparatus according to claim 1, further comprising a lid configured to be arranged to face a second surface opposite to the first surface of the cleaning object.

9. The cleaning apparatus according to claim 8, further comprising at least one third supply port, through which a fluid is supplied to the cleaning object, on a surface of the lid that faces the cleaning object.

10. A method for cleaning the cleaning object in the cleaning apparatus according to claim 1, comprising: a supporting step including supporting the cleaning object with the supporting portion; a rotating step including supplying the rotational fluid from the at least one first supply port to the cleaning object and causing the cleaning object supported by the supporting portion to rotate; and a cleaning step including cleaning the cleaning object while the cleaning object rotates.